

Section 4.1 Worksheet

Student work

1. Let $\mathbf{u} = (5, -2)$ and $\mathbf{v} = (-3, 7)$. Find $\mathbf{u} + \mathbf{v}$, $2\mathbf{u} - \mathbf{v}$, and $-\frac{1}{2}\mathbf{v}$.
2. Let $\mathbf{a} = (2, 1, -1)$, $\mathbf{b} = (0, 3, 2)$, and $\mathbf{c} = (1, -2, 4)$. Compute $3\mathbf{a} - 2\mathbf{b} + \mathbf{c}$.
3. Solve $2(\mathbf{x} - \mathbf{u}) = \mathbf{v} + \mathbf{x}$ when $\mathbf{u} = (1, 2, 3)$ and $\mathbf{v} = (4, -1, 2)$.
4. Determine c_1, c_2 such that $(9, 21) = c_1(1, 1) + c_2(1, -1)$.
5. A badminton player makes court movements $(2, 1)$, $(-1, 3)$, and $(0, -2)$. Find the net displacement.

Section 4.2 Worksheet

Student work

Determine whether each set is a vector space under the usual operations. Justify.

1. $\{(x, y, z) : x + y - z = 0\}$.
2. $\{(x, y) : x + y = 1\}$.
3. Polynomials of degree at most three.
4. 2×2 matrices with determinant 1.
5. Continuous functions satisfying $f(0) = 0$.
6. Explain why finitely many numerical tests cannot prove all vector-space axioms.

Section 4.3 Worksheet

Student work

Determine whether each is a subspace.

1. $\{(x, y, z) : 2x - y + 3z = 0\}$.
2. $\{(x, y, z) : 2x - y + 3z = 7\}$.
3. $\{p \in P_3 : p(0) = p(1) = 0\}$.
4. Symmetric 2×2 matrices.
5. Invertible 2×2 matrices.
6. Find a parametric description for $x + y - z = 0$.

Section 4.4 Worksheet

Student work

1. Is $(8, 1, 7)$ in the span of $(1, 2, 0)$, $(0, 1, 1)$, and $(2, 0, 1)$?
2. Is $\{(1, 0, 1), (0, 1, 1), (1, 1, 2)\}$ independent?
3. Does $\{(1, 2), (3, 6)\}$ span $\mathbb{R}^2$?
4. Test $\{1 + x, x + x^2, 1 + 2x + x^2\}$ in P_2 for independence.
5. Find a dependence relation among $(1, 0, 2)$, $(0, 1, 1)$, $(2, 1, 5)$.
6. Explain geometrically why two nonparallel vectors span $\mathbb{R}^2$.

Section 4.5 Worksheet

Student work

1. Is $\{(1, 0, 1), (0, 1, 1), (1, 1, 0)\}$ a basis for $\mathbb{R}^3$?
2. Find $\dim \text{span}\{(1, 2, 0), (0, 1, 1), (1, 3, 1)\}$.
3. Give a standard basis and dimension for P_4 .
4. Give a basis and dimension for symmetric 2×2 matrices.
5. Find a basis and dimension for $\{(s + t, s - t, 3s)\}$.
6. Why must five vectors in P_3 be dependent?

Section 4.6 Worksheet

Student work

Let $B = \begin{bmatrix} 1 & 2 & 0 & 1 \\ 2 & 4 & 1 & 3 \\ -1 & -2 & 1 & 0 \end{bmatrix}$.

1. Find the RREF and rank.
2. Find a row-space basis.
3. Find a column-space basis from the original matrix.
4. Find a nullspace basis and nullity.
5. Verify rank-nullity.
6. If $\mathbf{x}_p = (1, 0, 2, -1)^T$, find $\mathbf{b} = B\mathbf{x}_p$ and the complete solution of $B\mathbf{x} = \mathbf{b}$.

Section 4.7 Worksheet

Student work

1. For $B = \{(1, 2), (2, -1)\}$, find $[(8, 7)]_B$.
2. Reconstruct $\mathbf{x}$ from $[\mathbf{x}]_B = (3, -2)^T$.
3. For $B = \{(1, 1), (1, -1)\}$, find $P_{S \leftarrow B}$ and $P_{B \leftarrow S}$.
4. Convert $[\mathbf{x}]_B = (5, 4)^T$ to standard coordinates.
5. Find coordinates of $3 - 2x + 4x^2$ relative to $\{1, x, x^2\}$.
6. Find coordinates relative to $\{1, x - 1, (x - 1)^2\}$.

Section 4.8 Worksheet

Student work

1. Verify that $\cos x$ and $\sin x$ solve $y'' + y = 0$.
2. Compute their Wronskian.
3. Compute the Wronskian of $1, x, x^2$.
4. Identify center and semiaxes of $\frac{(x-3)^2}{16} + \frac{(y+1)^2}{9} = 1$.
5. Identify vertex and opening of $(y-2)^2 = -12(x+1)$.
6. Rotate $4x^2 + 4xy + 4y^2 = 12$ by 45° and classify.